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PAPER OF PT3-ACC-24CAI0104-B2024-SEM2 EXAM

Q1. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-ARRAY

Search an element in an array

You are given an array A

A of size N

N and an element X

X. Your task is to find whether the array A

A contains the element X

X or not.

Input Format

- The first line contains two space-separated integers N and X
- *where N* is the size of array and X is the element to be searched.

- The second line contains all the elements of array A

Output Format

Output "YES" if the element X is present in A

otherwise output "NO".

Constraints

- $1 \le N \le 10^5$
- $1 \le A_i \le 10^5$

Sample 1:

Input:

5 3

7 3 5 2 1

Output:

YES

Sample 2:

Input:

5 10

7 3 5 2 1

output:

```
NO
```

ANSWER:

Test case 1 :

Input:
5 3
7 4 5 2 3
Output:
YES

Test case 2 :

Input:
5 10
7 3 5 2 1
Output:
NO

Q2. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-ARRAY

Find maximum in an Array

Given a list of N integers, representing height of mountains. Find the height of the tallest mountain.

Input:

- First line will contain T, number of testcases. Then the testcases follow.
- The first line in each testcase contains one integer, N
- The following line contains *N* space separated integers: the height of each mountains.

Output:

For each test case, output one line with one integer: the height of the tallest mountain for that test case.

Constraints

- $1 \le T \le 10$
- $1 \le N \le 100000$
- $0 \le$ height of each mountain $\le 10^9$

Sample Input:

1

5

4 7 6 3 1

Sample Output:

7

ANSWER:

Test case 1 :

Input:
1
5
4 7 3 5 1
Output:
7

Test case 2 :

Input:
3
2
1 2
4
9 6 4 2
3
3 2 1
Output:
2
9
3

Q3. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-STRING

DNA Storage

For encoding an even-length binary string into a sequence of A, T, C, and G, we iterate from left to right and replace the characters as follows:

00 is replaced with A

01 is replaced with T

10 is replaced with C

11 is replaced with G

Given a binary string S

S of length N(where *N* is even),

find the encoded sequence.

Input Format

- First line will contain T, number of test cases. Then the test cases follow.
- Each test case contains two lines of input.

- First line contains a single integer *N*, the length of the sequence.
- Second line contains binary string S of length N
-

Output Format

For each test case, output in a single line, the encoded sequence.

Note: Output is **case-sensitive**.

Constraints

- 1≤T≤100
- 2≤N≤10^3
- *N* is even.
- Sum of N over all test cases is at most 10^3

S contains only characters 0 and 1.

Sample Input:

4

2

00

4

0011

6

101010

4

```
1001


Sample Output:

A

AG

CCC

CT
```

ANSWER:

Test case 1 :

Input:
4
2
00
4
0011
6
101010
4
1001
Output:
A
AG
CCC
CT

Q4. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-STRING

Wordle

Chef invented a modified wordle.

There is a hidden word S and a guess word T, both of length 5

Chef defines a string M to determine the correctness of the guess word. For the *ith* index:

- If the guess at the i*ith* index is correct, the ith character of M is G.
- If the guess at the ith index is wrong, the ith character of M is B

Given the hidden word S and guess T, determine string M

Input Format

- First line will contain T number of test cases. Then the test cases follow.
- Each test case contains of two lines of input.
- First line contains the string S, *S* - the hidden word.
- Second line contains the string T, *T* - the guess word.

Output Format

For each test case, print the value of string M.

You may print each character of the string in uppercase or lowercase (for example, the strings BgBgB

BgBgB, BGBGB

BGBGB, bgbGB

bgbGB and bgbgb

bgbgb will all be treated as identical).

Constraints

- $1 \leq T \leq 1000$
- $|S|=|T|=5$
- S,T contain uppercase English alphabets only.

Sample Input:

3

ABCDE

EDCBA

ROUND

RINGS

START

STUNT

Sample Output:

BBGBB

GBBBB

GGBBG

Explanation:

Test Case 1**:** Given string $S = \texttt{ABCDE}$ and $T = \texttt{EDCBA}$. The string M is:

- Comparing the first indices, $\texttt{A} \neq \texttt{E}$, thus, $M[1] = \texttt{B}$.
- Comparing the second indices, $\texttt{B} \neq \texttt{D}$, thus, $M[2] = \texttt{B}$.
- Comparing the third indices, $\texttt{C} = \texttt{C}$, thus, $M[3] = \texttt{G}$.
- Comparing the fourth indices, $\texttt{D} \neq \texttt{B}$, thus, $M[4] = \texttt{B}$.
- Comparing the fifth indices, $\texttt{E} \neq \texttt{A}$, thus, $M[5] = \texttt{B}$.

  Thus, $M = \texttt{BBGBB}$.

Test Case 2**:** Given string $S = \texttt{ROUND}$ and $T = \texttt{RINGS}$. The string M is:

- Comparing the first indices, $\texttt{R} = \texttt{R}$, thus, $M[1] = \texttt{G}$.
- Comparing the second indices, $\texttt{O} \neq \texttt{I}$, thus, $M[2] = \texttt{B}$.
- Comparing the third indices, $\texttt{U} \neq \texttt{N}$, thus, $M[3] = \texttt{B}$.
- Comparing the fourth indices, $\texttt{N} \neq \texttt{G}$, thus, $M[4] = \texttt{B}$.
- Comparing the fifth indices, $\texttt{D} \neq \texttt{S}$, thus, $M[5] = \texttt{B}$.

  Thus, $M = \texttt{GBBBB}$.

ANSWER:

Test case 1 :

Input:
3
ABCDE
EDCBA
ROUND
RINGS
START
STUNT
Output:
BBGBB
GBBBB
GGBBG

Q5. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-ARRAY

Take discount or Not

There are N items in a shop. You know that the price of the *i*-th item is Ai. Chef wants to buy all the N items.

There is also a discount coupon that costs X rupees and reduces the cost of every item by Y rupees. If the price of an item was initially ≤Y, it becomes free, i.e, costs 0.

Determine whether Chef should buy the discount coupon or not. Chef will buy the discount coupon if and only if the total price he pays after buying the discount coupon is **strictly less** than the price he pays without buying the discount coupon.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases. The description of the test cases follows.
- Each test case consists of two lines of input.
  - The first line of the test case contains three space-separated integers — N, X, and Y.
  - The second line contains N space-separated integers — $A_1, A_2, \ldots, A_N$.

Output Format

- For each test case, output `COUPON` if Chef should buy the discount coupon, and `NO COUPON` otherwise.

Each letter of the output may be printed in either lowercase or uppercase. For example, the strings `coupon`, `CouPoN`, and `COUPON` will all be treated as equivalent.

Constraints

- $1 \leq T \leq 1000$
- $1 \leq N \leq 100$
- $1 \leq X, Y \leq 10^5$
- $1 \leq A_i \leq 10^5$

Sample Input:

5

4 30 10

15 8 22 6

4 40 10

15 8 22 6

4 34 10

15 8 22 6

2 10 100

60 80

3 30 5

50 60 50

Sample Output:

COUPON

NO COUPON

NO COUPON

COUPON

NO COUPON

Explanation:

Test case 1: The original cost of the items is $15 + 8 + 22 + 6 = 51$. Buying the coupon costs 30, and after buying it the cost of buying all the items is $5 + 0 + 12 + 0 = 17$. The total cost of buying everything with the coupon is $30 + 17 = 47$, which is strictly less than 51. So, Chef will buy the coupon.

Test case 2: The original cost of the items is $15 + 8 + 22 + 6 = 51$. Buying the coupon costs 40, and after buying it the cost of buying all the items is $5 + 0 + 12 + 0 = 17$. The total cost of buying everything with the coupon is $40 + 17 = 57$, which is more than 51. So, Chef will not buy the coupon.

Test case 3: The original cost of the items is 51. Buying the coupon costs 34, and the cost of buying all the items after using it is 17, making the total cost $34 + 17 = 51$. Since this is not strictly less than the original cost, Chef won't buy the coupon.

Test case 4: The original cost of the items is 140, the coupon costs 10, and the cost of buying everything after using the coupon is 0. Since $10 + 0 < 140$, Chef will buy the coupon.

ANSWER:

Test case 1 :

Input:
5
4 30 10
15 8 22 6
4 40 10
15 8 22 6
4 34 10
15 8 22 6
2 10 100
60 80

3 30 5
50 60 50
Output:
COUPON
NO COUPON
NO COUPON
COUPON
NO COUPON

Q6. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-ARRAY

Cost of Groceries

Chef visited a grocery store for fresh supplies. There are N items in the store where the ith item has a freshness value Ai and cost Bi.

Chef has decided to purchase **all** the items having a freshness value **greater than equal to** X.

Find the total cost of the groceries Chef buys.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.
- The first line of each test case contains two space-separated integers *N* and X where N is the number of items and X is the minimum freshness value an item should have.
- The second line contains N space-separated integers, the array *A*, denoting the freshness value of each item.
- The third line contains N space-separated integers, the array B, denoting the cost of each item.

Output Format

For each test case, output on a new line, the total cost of the groceries Chef buys.

Constraints

- $1 \le T \le 100$
- $1 \le N, X \le 100$
- $1 \le A_i, B_i \le 100$

Sample Input:

4

2 20

15 67

10 90

3 1

1 2 3

1 2 3

3 100

10 90 50

30 7 93

4 50

12 78 50 40

40 30 20 10

Sample Output:

90

6

0

50

Explanation:

Test case 1**:** Item 2 has freshness value greater than equal to $X = 20$. Thus, Chef buys item 2. The total cost is 90.

Test case 2**:** Items $1, 2,$ and 3 have freshness value greater than equal to $X = 1$. Thus, Chef buys all 3 items. The total cost is $1 + 2 + 3 = 6$.

Test case 3**:** No item has freshness value greater than equal to $X = 100$. Thus, Chef buys no items.

Test case 4**:** Items 2 and 3 have freshness value greater than equal to $X = 50$. Thus, Chef buys items 2 and 3. The total cost is $30 + 20 = 50$.

ANSWER:

Test case 1 :

Input:
4
2 20
15 67
10 90
3 1

1 2 3
1 2 3
3 100
10 90 50
30 7 93
4 50
12 78 50 40
40 30 20 10
Output:
90
6
0
50

Q7. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-ARRAY

MIN To MAX

You are given an array A of size N.

Let M be the **minimum** value present in the array initially.

In one operation, you can choose an element $Ai(1 \leq i \leq N)$ and an integer X $(1 \leq X \leq 100)$, and set Ai=X.

Determine the **minimum** number of operations required to make M the **maximum** value in the array A.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.
- The first line of each test case contains a single integer N, where N - the size of the array.
- The next line of each test case contains N space-separated integers A1,A2,…,AN - the elements of the array.

Output Format

For each test case, output on a new line, the **minimum** number of operations required to make M, the **maximum** value in the array A.

Constraints

- $1 \leq T \leq 100$
- $1 \leq N \leq 100$
- $1 \leq A_i \leq 100$

Sample Input:

3

2

1 2

4

2 2 3 4

1

1

Sample Output:

1

2

0

Explanation:

Test case 1: The minimum value in the array, M, is initially 1. We can use one operation as following:

- Choose A_2 and set it as $X = 1$. Thus, the final array becomes $[1, 1]$.

Since all elements of the final array are 1, the maximum value of the array is now 1. It can be shown that this is the minimum number of operations required to do so.

Test case 2: The minimum value in the array, M, is initially 2. We can use two operations as following:

- Choose A_4 and set it as $X = 2$. Thus, the array becomes $[2, 2, 3, 2]$.
- Choose A_3 and set it as $X = 2$. Thus, the array becomes $[2, 2, 2, 2]$.

Since all elements of the final array are 2, the maximum value of the array is now 2.

Test case 3: The minimum value in the array, M, is initially 1. It is also the maximum value of the array. Hence, no operations are required.

ANSWER:

Test case 1 :

Input:
3
2
1 2
4
2 2 3 4
1
1
Output:
1
2
0

Q8. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-STRING

Different Consecutive Characters

Chef has a binary string S

S of length N. Chef can perform the following operation on S:

- Insert any character (0 or 1) at any position in S.

Find the minimum number of operations the Chef needs to perform so that no two consecutive characters are the same in S

Input Format

- The first line contains a single integer T.
- *T* — the number of test cases. Then the test cases follow.
- The first line of each test case contains an integer N
- *N* — the length of the binary string S.
- The second line of each test case contains a binary string S of length N containing 0s and 1s only.

Output Format

For each test case, output on a new line the minimum number of operations Chef needs to perform so that no two consecutive characters are the same in S.

Constraints

- $1 \le T \le 100$
- $1 \le N \le 1000$

Sample Input:

3

2

11

4

0101

5

00100

Sample Output:

1

0

2

Explanation:

Test case 1: We can perform the following operations: $11 \rightarrow 1\underline{0}1$.

Test case 2: We do not need to perform any operations.

Test case 3: We can perform the following operations: $00100 \rightarrow 0\underline{1}0100 \rightarrow 0101\underline{0}10$.

ANSWER:

Test case 1 :

Input:
3
2
11
4
0101
5
00100
Output:
1
0
2

Q9. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-STRING

Convert String to Title Case

Given a string S consisting of only lowercase and uppercase English letters and spaces, your task is to convert it into title case. In title case, the first letter of each word is capitalized while the rest are in lowercase, except for words that are entirely in uppercase (considered as acronyms), which should remain unchanged.

Note:

- Words are defined as contiguous sequences of English letters separated by spaces.
- Acronyms are words that are entirely in uppercase and should remain unchanged.
- Assume the input does not contain leading, trailing, or multiple spaces between words.
-

Input Format

- The first line contains a single integer T, the number of test cases.
- Each of the next T lines contains a string S.
-

Output Format

For each test case, print a single line containing the string S converted into title case.

Constraints

- $1 \le T \le 100$
- $1 \le |S| \le 1000$, where $|S|$ is the length of the string.

Sample Input:

5

hello world

this is a CODECHEF problem

WELCOME to the JUNGLE

the quick BROWN fOx

programming in PYTHON

Sample Output:

Hello World

This Is A CODECHEF Problem

WELCOME To The JUNGLE

The Quick BROWN Fox

Programming In PYTHON

Explanation:

- In the first test case, each word is capitalized as they are not acronyms.
- In the second test case, "CODECHEF" is an acronym and remains in uppercase.
- In the third test case, "WELCOME" and "JUNGLE" are considered acronyms.
- In the fourth test case, "BROWN" is an acronym, while the rest of the words follow the title case rule.
- In the fifth test case, "PYTHON" is an acronym, and the rest of the string is converted to title case.

ANSWER:

Test case 1 :

Input:
5
hello world
this is a CODECHEF problem
WELCOME to the JUNGLE
the quick BROWN fOx
programming in PYTHON
Output:

Hello World
This Is A CODECHEF Problem
WELCOME To The JUNGLE
The Quick BROWN Fox
Programming In PYTHON

Q10. ACC-24CAI0104-CODING-5MARKS-PT3-B2024-AI-STRING

World Chess Championship

The World Chess Championship 2022 is about to start. 14 Classical games will be played between Chef and Carlsen in the championship, where each game has one of three outcomes — it can be won by Carlsen, won by Chef, or it can be a draw. The winner of a game gets 2 points, and the loser gets 0 points. If it's a draw, both players get 1 point each.

The total prize pool of the championship is 100* X. At end of the 14 Classical games, if one player has **strictly more** points than the other, he is declared the champion and gets 60*X as his prize money, and the loser gets 40*X.

If the total points are **tied**, then the defending champion Carlsen is declared the winner. However, if this happens, the winner gets only 55*X and the loser gets 45*X.

Given the results of all the 14 games, output the prize money that Carlsen receives.

The results are given as a string of length 14

14 consisting of the characters C, N, and D.

C denotes a victory by Carlsen.

N denotes a victory by Chef.

D denotes a draw.

Input Format

- The first line of input contains an integer *T*, denoting the number of test cases. The description of T test cases follows.
- The first line of each test case contains one integer X, denoting that the total prize pool is 100*X
- The second line contains the results of the match, as a string of length 14 containing only the characters C, N, and D.

Output Format

For each test case, output in a single line the total prize money won by Carlsen.

Constraints

- 1≤T≤1000
- 1≤X≤10^6
- |S|=14
- S contains only characters D, C, N.

Sample Input:

4

100

CCCCCCCCCCCCCC

400

CDCDCDCDCDCDCD

30

DDCCNNDDDCCNND

1

NNDNNDDDNNDNDN

Sample Output:

6000

24000

1650

40

Explanation:

Test case 1: Since Carlsen won all the games, he will be crowned the champion and will get $60 \cdot X$ as the prize money which is $60 \cdot 100 = 6000$

Test case 2: Carlsen won 7 games and drew 7, so his score is $2 \cdot 7 + 1 \cdot 7 = 21$. Chef lost 7 games and drew 7, so his score is $0 \cdot 7 + 1 \cdot 7 = 7$. Since Carlsen has more points, he will be crowned the champion and will get $60 \cdot X$ as the prize money which is $60 \cdot 400 = 24000$

Test case 3: Carlsen and Chef both end up with 14 points. So, Carlsen is declared the winner, but because the points were tied, he receives $55 \cdot X = 55 \cdot 30 = 1650$ in prize money.

ANSWER:

Test case 1 :

Input:
4
100
CCCCCCCCCCCCCC
400
CDCDCDCDCDCDCD
30

DDCCNNDDDCCNND
1
NNDNNDDDNNDNDN
Output:
6000
24000
1650
40